

Fractal-Wave Algebra: Fundamental Ontology

1. Ontological Starting Point

FWA describes reality as dynamically generated structures rather than static objects.

$I \rightarrow S \rightarrow 0' \rightarrow \Phi \rightarrow w \rightarrow R \rightarrow O$

I = information

S = symmetry

$0'$ = dynamical zero

Φ = field state

w = wave realization

R = resonance

O = observable structure

Central principle:

process precedes object

2. Information and Symmetry

$I = dS$

$L_u = S_{ideal} - S_{real}$

symmetry $\rightarrow$ distinction $\rightarrow$ information $\rightarrow$ dynamics

3. Dynamical Zero

$0' \neq 0$

$0' = 0 + d$

4. Field as Process

$\Phi = \Phi(r, t)$

$\Phi(n+1) = F(\Phi(n), \text{grad}(\Phi(n)), d\Phi/dt, T_n)$

5. Wave as Manifestation

$w = f(0')$

6. Resonance and Stable Structure

$\Phi \rightarrow$ resonance $\rightarrow$ stable mode

particle = stable fractal-wave mode

7. Derived Reality

information $\rightarrow$ symmetry breaking $\rightarrow$ field dynamics $\rightarrow$ wave $\rightarrow$ resonance $\rightarrow$ stable structure $\rightarrow$ observable

8. Fundamental Principle

Reality is dynamically generated structure.

$$O = M[R(F(I,S,O',\Phi))]$$